7A INTERMOLECULAR INTERACTIONS

SPECIFICATION REFERENCE 7.1 7.2 7.4

LEARNING OBJECTIVES

- Understand the nature of intermolecular forces resulting from the following interactions:
  (i) London forces
  (ii) permanent dipoles
  (iii) hydrogen bonds.
- Understand the interactions in molecules, such as H_2O, liquid NH_3 and liquid HF, which give rise to hydrogen bonding.
- Predict the presence of hydrogen bonding in molecules analogous to those mentioned above.

BACKGROUND TO NON-BONDED INTERMOLECULAR INTERACTIONS

A number of interactions between molecules are considerably weaker than typical covalent and polar covalent bonds. These interactions are usually described as:

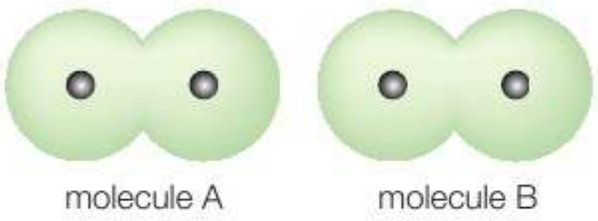
- 'non-bonded interactions' or
- 'intermolecular' because they occur *between* molecules.

The most important non-bonded interactions are 'London forces'. They have this name because their existence was first suggested in 1930 by Fritz London, a German physicist. London forces are also sometimes referred to as 'dispersion forces'. Although they are weaker than covalent and polar covalent bonds, London forces play an important part in determining the physical and chemical properties of many molecules.

Other intermolecular interactions arise from the permanent dipoles that exist in some molecules.

LONDON FORCES

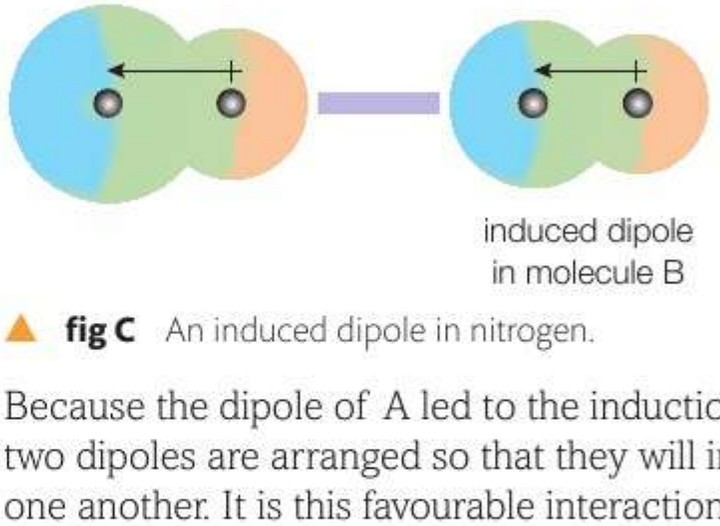
We can describe this interaction by considering two non-polar molecules of nitrogen, labelled A and B in **fig A**.



▲ **fig A** Electron density in nitrogen.

Each molecule is non-polar because, on average, the electron density is symmetrically distributed throughout the molecule (see **Topic 3**).

However, electron density fluctuates over time. If, at any time, the electron density becomes unsymmetrical in molecule A, a dipole will be generated, as shown in **fig B**.



▲ **fig B** An instantaneous dipole in nitrogen.

The electron density on the left of the molecule has increased, giving that end of the molecule a partial negative charge ($\delta-$). However, the electron density on the right has decreased. This gives that end of the molecule a partial positive charge ($\delta+$). For this reason, an instantaneous dipole is created in molecule A.

The $\delta+$ end of molecule A is closer to molecule B, so the electron density of molecule B is pulled to the left. This generates a partial negative charge on the left-hand end of the molecule, and a partial positive charge on the right-hand end. This creates an induced dipole in molecule B, as shown in **fig C**.

▲ **fig C** An induced dipole in nitrogen.

Because the dipole of A led to the induction of the dipole in B, the two dipoles are arranged so that they will interact favourably with one another. It is this favourable interaction that is responsible for the London force of attraction between the two molecules.

It is important to realise that the induced dipole will *always* be aligned in such a way that the interaction with the instantaneous dipole is favourable. The fluctuations that lead to the generation of an instantaneous dipole, and the subsequent induction of a dipole in a nearby molecule, are very rapid processes. This is especially the case when they are compared with the rates at which the molecules are moving. This is because of their kinetic energy and rotational energy. As the two molecules move around, they will continue to attract each other regardless of their orientation.

EXAM HINT

If you are asked to explain London forces in an exam question, it is always worth drawing a diagram and then referring to it in your answer. A good diagram is often better than just a written answer.

A feature of London forces is that the attractive force increases with increasing number of electrons in the molecule. We can demonstrate this by observing the boiling temperatures of the noble gases (**table A**). The noble gases all exist as monatomic molecules (i.e. single atom molecules). The London force is the only force of attraction between the molecules. The stronger the force of attraction, the more energy is required to separate the molecules. The boiling temperature increases because of this.

GAS	HELIUM	NEON	ARGON	KRYPTON	XENON	RADON
Boiling temperature / K	4.3	27.1	87.4	121	165	211

table A Boiling temperatures of the noble gases.

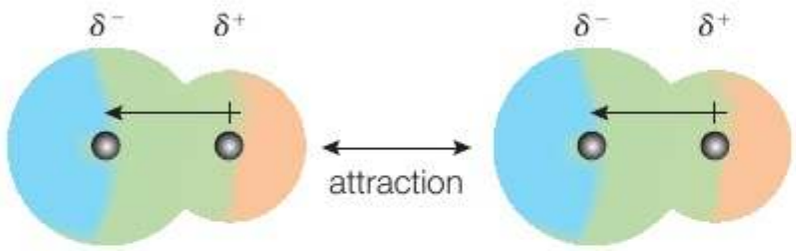
The more electrons there are in a molecule, the greater the fluctuation in electron density and the larger the instantaneous and induced dipoles created. There is a similar trend in the boiling temperatures of the halogens from F_2 to I_2 (see **Topic 8**).

A second feature of London forces is that they depend on the shape and size of the molecules. The more points of contact there are between the molecules, the greater the overall London force. We will look at this in more detail in **Topic 7B**.

A third feature is that London forces are always present between molecules. This is the case whether or not they have a permanent dipole and whether or not they form hydrogen bonds with each other (see below).

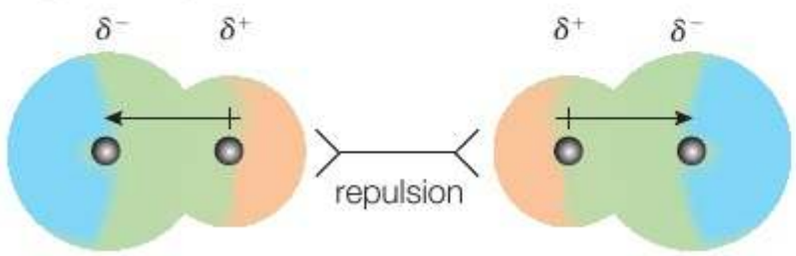
PERMANENT DIPOLES

If the molecules possess permanent dipoles, they will also interact with one another. If the dipoles are aligned correctly, then there will be a favourable interaction and the two molecules will attract one another (**fig D**).



▲ **fig D** Attraction between permanent dipoles.

The problem here is that, for example, in a liquid, the random movement of the molecules is such that the dipoles are not always aligned to produce a favourable interaction.



▲ **fig E** Repulsion between permanent dipoles.

As a result, when averaged out, the interaction between permanent dipoles is usually much less than the interaction between instantaneous and induced dipoles. The London force is usually the more significant interaction between molecules.

LEARNING TIP

Initially, it seems surprising that the interaction between permanent dipoles should be weaker than the interaction between instantaneous and induced dipoles.

The key point is that induced dipoles are always aligned so that their interaction is favourable. This is not true for permanent dipoles.

It is possible for a molecule with a permanent dipole to induce a dipole in a nearby molecule. The two types of interaction, permanent dipole–permanent dipole and permanent dipole–induced dipole, are sometimes put together under the heading permanent dipole–dipole forces.

SUMMARY

Table B is a useful reminder of the origins of non-bonded intermolecular interactions.

NAME OF INTERACTION	ORIGIN
London forces	instantaneous dipole-induced dipole interaction
Permanent dipoles	permanent dipole-permanent dipole interaction

table B Summary of non-bonded intermolecular interactions.

LEARNING TIP

It is important to realise that London forces exist between all types of molecules, whether non-polar or polar.

For example, London forces, as well as permanent dipole-permanent dipole interactions, exist between the polar molecules of hydrogen chloride.

DID YOU KNOW?

The term 'van der Waals force' is often used in conjunction with intermolecular interactions.
The van der Waals force is the sum of all the intermolecular interactions between the molecules. The term includes London forces and permanent dipole-permanent dipole interactions.

EXAM HINT

It is better *not* to use the term van der Waals forces in your written answers and use only London forces, and permanent dipole-permanent dipole interactions.

THE HYDROGEN BOND

There is one other intermolecular interaction that is, in some cases, very important. It is called a **hydrogen bond**.

The key to understanding the nature of the hydrogen bond is appreciating that the atom bonded to hydrogen has to be more electronegative than hydrogen, and that there must be some evidence of bond formation between the hydrogen and another atom, either within the same molecule ('intramolecular hydrogen bonding') or in a different molecule ('intermolecular hydrogen bonding').

DID YOU KNOW?

We are only looking at *inter*molecular hydrogen bonding, although *intra*molecular hydrogen bonding plays a significant role in many biochemical molecules such as DNA and proteins.
Because there has to be evidence of some bond formation (the nature of the evidence, largely from various types of spectroscopy, is beyond this course), there is an argument that hydrogen bonds could be considered to be bonded intermolecular interactions. However, most academic texts still prefer to use the term non-bonded intermolecular interactions to describe them.

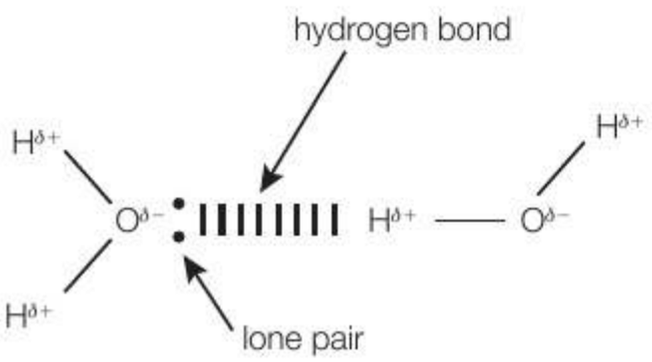
Hydrogen bonding is most significant when hydrogen is bonded to very small, highly electronegative atoms such as oxygen, nitrogen and fluorine, although it is not confined to these atoms.

HYDROGEN BONDING THROUGH OXYGEN

All compounds containing an –O—H group form intermolecular hydrogen bonds.

The most important example is water.

The hydrogen bond forms between the oxygen atom of one water molecule and the hydrogen atom of a second water molecule (**fig F**).



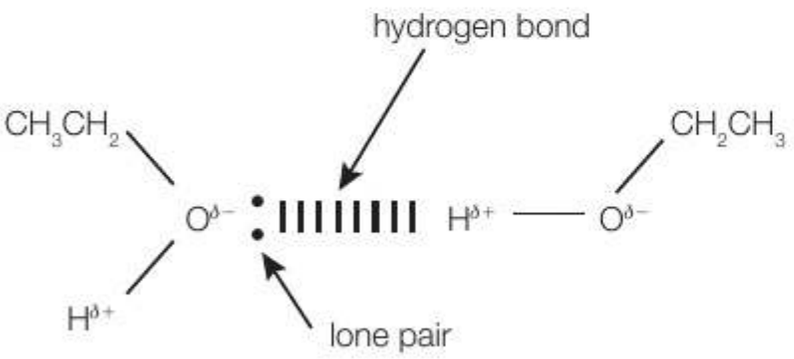
▲ **fig F** Hydrogen bonding between water molecules.

The interaction is not just that of an extreme dipole–dipole interaction. There is some partial bond formation using a lone pair of electrons on the oxygen atom. Because the oxygen atom has two lone pairs, it can form hydrogen bonds with two other water molecules.

Another feature of a hydrogen bond that indicates partial bond formation is that, like covalent bonds, hydrogen bonds are directional in nature. The bond angle between the three atoms involved is often 180°, or close to it, but this is not always the case.

Alcohols (see **Topic 10**) also form intermolecular hydrogen bonds.

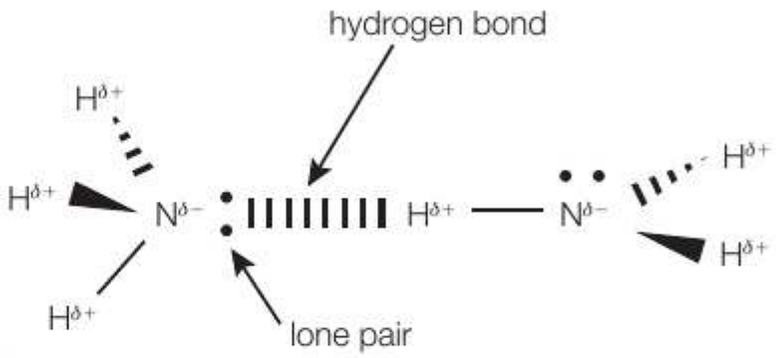
Fig G shows the formation of a hydrogen bond between two ethanol molecules.



▲ **fig G** Hydrogen bonding between ethanol molecules.

HYDROGEN BONDING THROUGH NITROGEN

All compounds containing an –N—H group can form intermolecular hydrogen bonds. An example is the organic group of compounds known as primary amines, which have the general formula RNH_2. **Fig H** shows the hydrogen bonding in ammonia, NH_3.



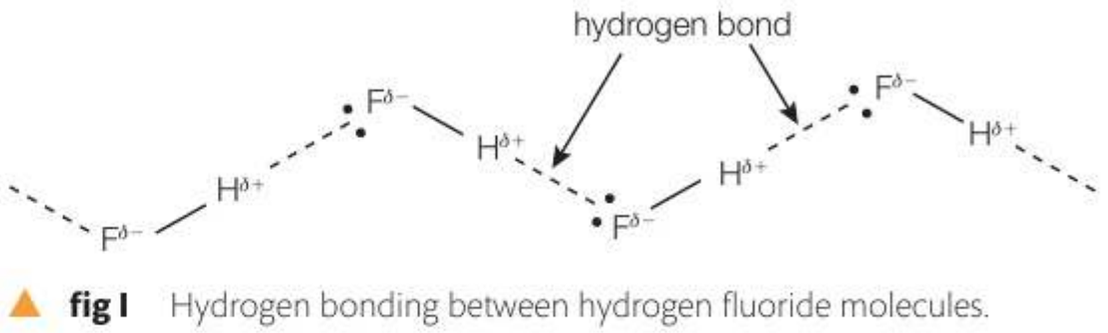
▲ **fig H** Hydrogen bonding between ammonia molecules.

DID YOU KNOW?

Hydrogen bonds are often quoted as being the strongest of all intermolecular interactions. This is true for some molecules such as water, but it is not true for the majority of molecules containing hydrogen bonds.
For example, although the hydrogen bond is significant in the case of the short chain alcohols, e.g. methanol (CH_3OH) and ethanol (CH_3CH_2OH), it becomes less significant for longer chain alcohols such as pentan-1-ol ($CH_3CH_2CH_2CH_2CH_2OH$) (see **Topic 7B**). The same trend is exhibited by the amines (see the section on amines in **Topic 19 (Book 2: IAL)**.
For this reason, it is not correct to state that the hydrogen bond is always the most significant intermolecular interaction in any given group of analogous compounds. In fact, we will see that some hydrogen bonds are very weak.

HYDROGEN BONDING THROUGH FLUORINE

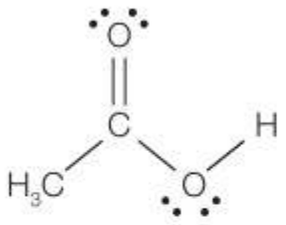
The only fluorine compound with intermolecular hydrogen bonding is hydrogen fluoride.



▲ **fig I** Hydrogen bonding between hydrogen fluoride molecules.

CHECKPOINT

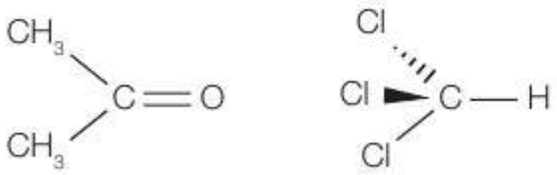
1. Molecules of ethanoic acid dimerise through hydrogen bonding when dissolved in certain organic solvents. The structure and shape of an ethanoic acid molecule is:



Draw a diagram to show how hydrogen bonds are formed between two molecules of ethanoic acid.

2. Explain how it is possible for a hydrogen bond to form between a molecule of propanone (CH_3COCH_3) and a molecule of trichloromethane ($CHCl_3$).

The structures and shapes of the molecules are:



3. The boiling temperature of ethanol (CH_3CH_2OH, 78.5 °C) is considerably higher than the boiling temperature of methoxymethane (CH_3OCH_3, −24.8 °C). The structures and shapes of the molecules are:



ethanol methoxymethane

In terms of the intermolecular interactions involved, explain the difference in boiling temperatures.

SUBJECT VOCABULARY

hydrogen bond an intermolecular interaction (in which there is some evidence of bond formation) between a hydrogen atom of a molecule (or molecular fragment) bonded to an atom which is more electronegative than hydrogen and another atom in the same or a different molecule

ADDITIONAL READING

Strong and weak hydrogen bonds

It was once believed that hydrogen bonds were formed only when hydrogen was bonded to oxygen, nitrogen or fluorine. Richard Nelmes of Edinburgh University surprised the world of chemistry when he discovered that solid hydrogen sulfide had an extended system of hydrogen bonding, which was similar to that in ice.

Because sulfur has a low electronegativity, the hydrogen bond between H_2S molecules is considered to be weak. In fact, it has a magnitude of 7 kJ mol^{-1}, compared with 22 kJ mol^{-1} in ice.

Some strengths of hydrogen bonds are shown in **table C** below. For comparison, the strengths of the 'full' bonds in some of these species are also shown.

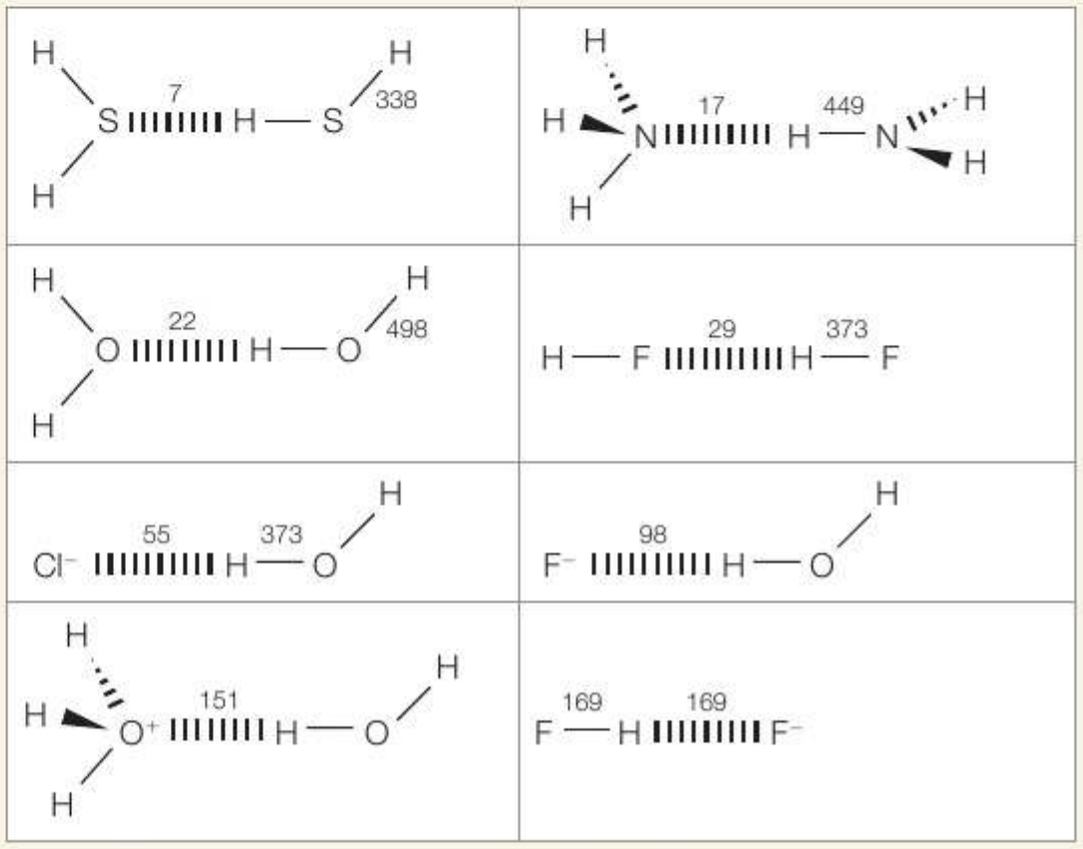


table C Strengths of some hydrogen bonds.

You will notice that, in most cases, the hydrogen bonds are very much weaker than the full bonds. You can also see that the strengths of the hydrogen bonds increase with the electronegativity of the element to which the hydrogen is attached. Hydrogen bonding strength increases as follows:

$$H_2S < NH_3 < H_2O < HF$$

Also note that the hydrogen bond to an ion is much stronger than that to a neutral molecule.

The strongest hydrogen bond shown in the table is that between HF and F^-. In this species the two bond lengths and bond strengths are identical. The resulting ion, $[F{-}H{-}F]^-$, is so stable that it is obtainable as the solid sodium salt, sodium hydrogen difluoride, $NaHF_2$.